# Appendix – Summary of RAG results by scenario, embedding model, and LLM

## Content and methodology

This appendix brings together the evaluation results of the question-answering system (RAG - Retrieval-Augmented Generation) built on a corpus of 140 multiple-choice questions. Each question is submitted to the full pipeline (retrieval of relevant passages, then answer generation by a large language model), and the generated answer is compared to the expected correct answer to compute an accuracy rate, expressed as the percentage of correct answers out of the 140 questions.

Three axes of variation were tested:

- The embedding model used to retrieve relevant passages (11 models compared: MiniLM-L12, AraModernBert-STS, MarBERTv2, Multilingual-E5, Arabic-Triplet-Matryoshka-V2, Arabic-SBERT-100K, AraBERTv2, CAMeLBERT, DistilBERT_Arabic, AraBERT_Large, and IslamQA-BGE-M3).
- The large language model (LLM) responsible for generating the final answer from the retrieved passages (Allam, Fanar, Llama3).
- The presence and type of reranking applied after the initial retrieval, before answer generation.

Two configurations are therefore distinguished: a "no reranking" configuration, where the passages retrieved by the embedding model are passed as-is to the LLM (Table A.1), and "with reranking" configurations, where an additional model reorders the passages before generation, tested solely to assess the contribution of this step. Five reranking scenarios were evaluated for each LLM: a Cross-Encoder MS-MARCO, a miniReranker Arabic v1, an ARA-Reranker V1, and two dual configurations chaining the miniReranker with one of the other two rerankers (Tables B.1, C.1, and D.1, for Allam, Fanar, and Llama3 respectively).

### Table A.1 — Accuracy by embedding model and by LLM (no reranking)

| Embedding model | Allam | Fanar | Llama3 |
|---|---|---|---|
| MiniLM-L12 | 87.14% | 84.29% | 72.86% |
| AraModernBert-STS | 83.57% | 81.43% | 80.00% |
| MarBERTv2 | 82.86% | 80.71% | 76.43% |
| Multilingual-E5 | 85.00% | 84.29% | 77.86% |
| Arabic-Triplet-Matryoshka-V2 | 83.57% | 84.29% | 77.86% |
| Arabic-SBERT-100K | 85.71% | 81.43% | 77.86% |
| AraBERTv2 | 84.29% | 79.29% | 79.29% |
| CAMeLBERT | 85.00% | 82.14% | 82.14% |
| DistilBERT_Arabic | 84.29% | 81.43% | 72.86% |
| AraBERT_Large | 83.57% | 82.86% | 65.00% |
| IslamQA-BGE-M3 | 82.86% | 81.43% | 78.57% |

*Table A.1: Accuracy (%) by embedding model for each LLM, in the no-reranking configuration. Models listed in the order they appear in the source results.*

Without reranking, Allam achieves the best results for nearly all embedding models, followed by Fanar, with Llama3 clearly trailing behind. The MiniLM-L12 embedding model achieves the best accuracy with Allam (87.14%), while CAMeLBERT stands out for the consistency of its performance across the three LLMs (82.14% to 85.00%). Conversely, AraBERT_Large shows the lowest score with Llama3 (65.00%), revealing a stronger sensitivity of this embedding model to the choice of generator LLM.

## Table B.1 — Accuracy by embedding model and by reranking scenario (LLM Allam)

| Embedding model | Cross-Encoder MS-MARCO | miniReranker Arabic v1 | ARA-Reranker V1 | Dual: miniReranker → Cross-Encoder | Dual: miniReranker → ARA-Reranker |
|---|---|---|---|---|---|
| MiniLM-L12 | 83.57% | 82.14% | 87.14% | 85.00% | 85.00% |
| AraModernBert-STS | 84.29% | 82.86% | 85.00% | 84.29% | 82.86% |
| MarBERTv2 | 83.57% | 83.57% | 85.00% | 82.86% | 84.29% |
| Multilingual-E5 | 82.86% | 83.57% | 85.00% | 84.29% | 85.71% |
| Arabic-Triplet-Matryoshka-V2 | 83.57% | 82.14% | 85.71% | 81.43% | 83.57% |
| Arabic-SBERT-100K | 84.29% | 80.71% | 84.29% | 82.86% | 82.14% |
| AraBERTv2 | 87.14% | 85.00% | 85.71% | 87.86% | 85.00% |
| CAMeLBERT | 87.86% | 89.29% | 86.43% | 84.29% | 85.71% |
| DistilBERT_Arabic | 82.86% | 79.29% | 85.00% | 83.57% | 82.14% |
| AraBERT_Large | 85.00% | 85.00% | 86.43% | 85.71% | 86.43% |
| IslamQA-BGE-M3 | 85.00% | 81.43% | 85.00% | 87.86% | 82.86% |

*Table B.1: Accuracy (%) by embedding model for each reranking scenario, with the LLM Allam.*

With Allam, several model/scenario pairs stand out clearly: CAMeLBERT combined with miniReranker Arabic v1 reaches 89.29%, the best accuracy across the whole appendix, and the ARA-Reranker V1 scenario performs strongly across most embedding models.

## Table C.1 — Accuracy by embedding model and by reranking scenario (LLM Fanar)

| Embedding model | Cross-Encoder MS-MARCO | miniReranker Arabic v1 | ARA-Reranker V1 | Dual: miniReranker → Cross-Encoder | Dual: miniReranker → ARA-Reranker |
|---|---|---|---|---|---|
| MiniLM-L12 | 81.43% | 79.29% | 79.29% | 80.71% | 84.29% |
| AraModernBert-STS | 80.00% | 82.86% | 84.29% | 77.14% | 79.29% |
| MarBERTv2 | 79.29% | 84.29% | 83.57% | 80.00% | 83.57% |
| Multilingual-E5 | 78.57% | 81.43% | 80.71% | 80.00% | 80.00% |
| Arabic-Triplet-Matryoshka-V2 | 76.43% | 81.43% | 85.00% | 75.71% | 82.86% |
| Arabic-SBERT-100K | 80.00% | 83.57% | 82.14% | 82.14% | 77.86% |
| AraBERTv2 | 75.71% | 82.14% | 80.00% | 83.57% | 85.00% |
| CAMeLBERT | 78.57% | 87.14% | 83.57% | 82.86% | 82.14% |
| DistilBERT_Arabic | 75.71% | 80.71% | 80.00% | 80.71% | 82.14% |
| AraBERT_Large | 77.86% | 80.00% | 80.00% | 77.86% | 80.00% |
| IslamQA-BGE-M3 | 80.71% | 81.43% | 80.71% | 80.71% | 83.57% |

*Table C.1: Accuracy (%) by embedding model for each reranking scenario, with the LLM Fanar.*

For Fanar, the effect of reranking is uneven across scenarios: the Cross-Encoder MS-MARCO tends to lower performance for most embedding models, while the miniReranker Arabic v1 and the Dual miniReranker → ARA-Reranker configuration tend to hold up well or improve results slightly. CAMeLBERT combined with miniReranker Arabic v1 remains the best combination observed for this LLM (87.14%).

## Table D.1 — Accuracy by embedding model and by reranking scenario (LLM Llama3)

| Embedding model | Cross-Encoder MS-MARCO | miniReranker Arabic v1 | ARA-Reranker V1 | Dual: miniReranker → Cross-Encoder | Dual: miniReranker → ARA-Reranker |
|---|---|---|---|---|---|
| MiniLM-L12 | 73.57% | 78.57% | 76.43% | 69.29% | 77.86% |
| AraModernBert-STS | 75.00% | 77.14% | 74.29% | 72.14% | 73.57% |
| MarBERTv2 | 77.14% | 76.43% | 80.00% | 72.14% | 80.71% |
| Multilingual-E5 | 70.71% | 78.57% | 78.57% | 75.00% | 80.00% |
| Arabic-Triplet-Matryoshka-V2 | 76.43% | 77.14% | 78.57% | 73.57% | 79.29% |
| Arabic-SBERT-100K | 76.43% | 75.71% | 77.86% | 73.57% | 81.43% |
| AraBERTv2 | 78.57% | 75.71% | 81.43% | 76.43% | 81.43% |
| CAMeLBERT | 75.71% | 81.43% | 75.71% | 75.00% | 82.86% |
| DistilBERT_Arabic | 71.43% | 73.57% | 73.57% | 72.86% | 71.43% |
| AraBERT_Large | 65.71% | 71.43% | 76.43% | 72.14% | 73.57% |
| IslamQA-BGE-M3 | 72.86% | 75.71% | 80.71% | 73.57% | 81.43% |

*Table D.1: Accuracy (%) by embedding model for each reranking scenario, with the LLM Llama3.*

Llama3 remains the weakest-performing LLM overall, with or without reranking. The dual miniReranker → ARA-Reranker configuration is nonetheless the most favorable for this LLM, reaching up to 82.86% with CAMeLBERT, while the dual miniReranker → Cross-Encoder configuration is consistently the weakest scenario for this LLM.